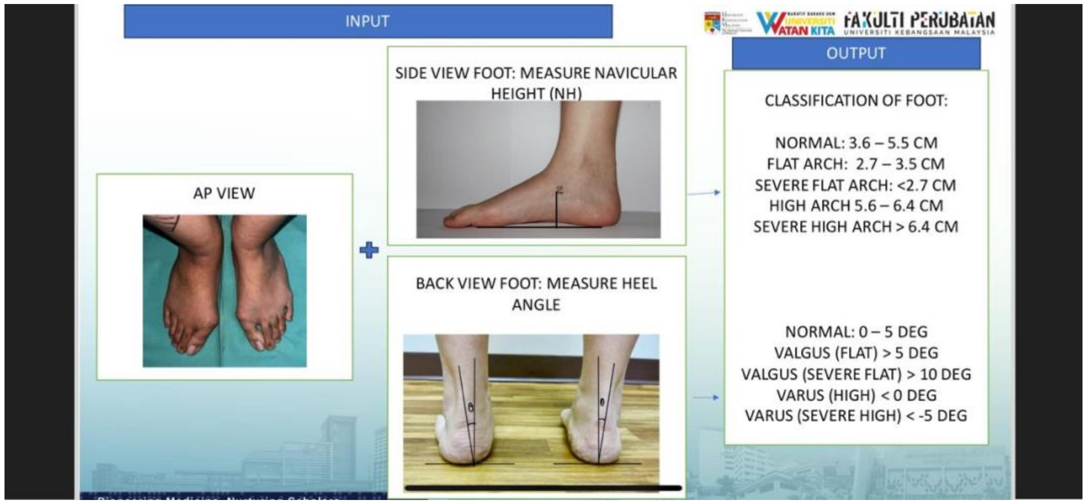
Insole Foot Classification

Project Overview

The system is designed to classify a user's foot characteristics using foot images and measurement data, then recommend or generate suitable insole configurations.



INPUT

Top/AP View

- Kite angle

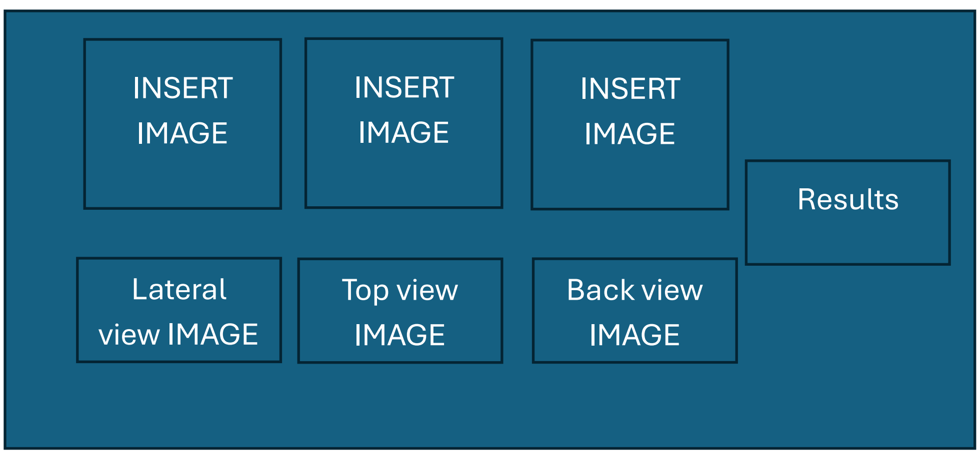
Lateral/Side View

- Calcaneal inclination
- Arch height
- 1st metatarsal–talus angle

Back/Posterior View

- Heel angle

EXPECTED OUTPUT (in exe file)



Classification (accuracy above 90% for unseen images)

Examples:

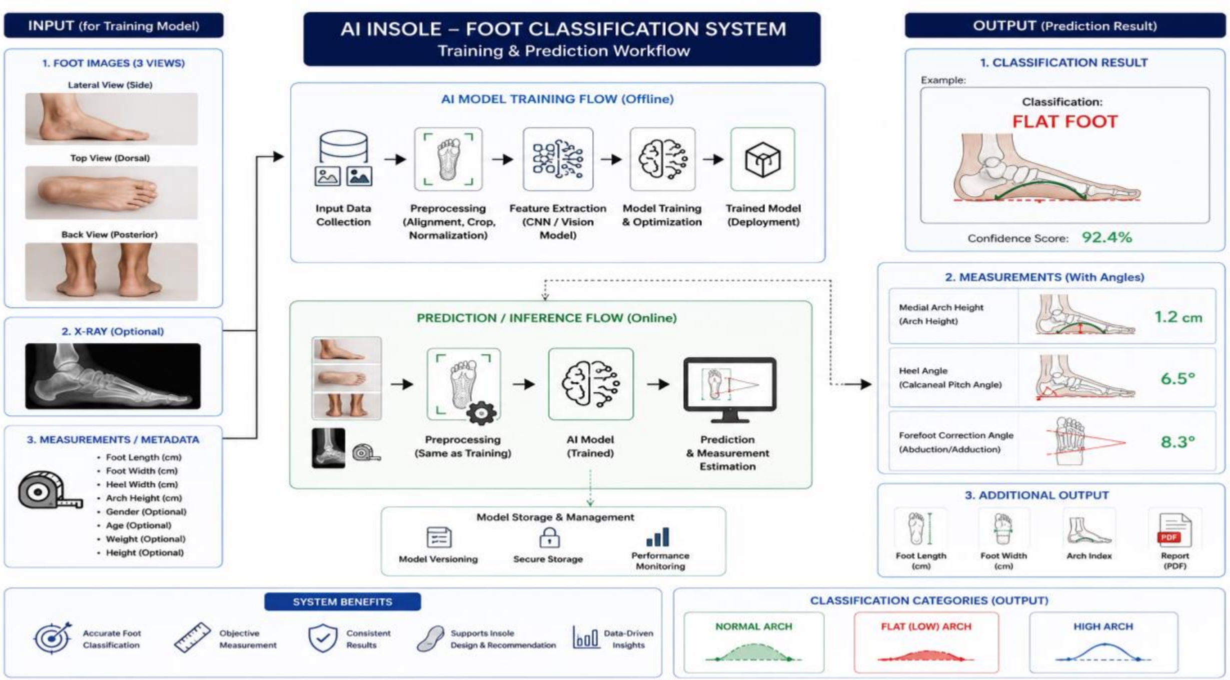
- Normal Foot
- Flat Arch
- Severe Flat Arch
- High Arch
- Severe High Arch

Measurements

- Calcaneal inclination: XX°
- Heel angle: XX°
- Arch height: XX cm
- Kite angle: XX°
- 1st metatarsal–talus angle: XX°

Description of Dataset

Measurement	Common View	Description
Calcaneal Inclination Angle	**Lateral View (Side/X-ray side)**	Measures the inclination/slope of the calcaneus (heel bone). Commonly used for arch assessment.
Heel Angle	**Back View (Posterior/AP rear view)**	Measures heel valgus/varus alignment from the rear of the foot. Often called rearfoot angle or calcaneal angle clinically.
Arch Height	**Lateral View (Side view)**	Measures medial longitudinal arch height (e.g., navicular height).
Kite Angle	**Top View / AP X-ray View**	Talocalcaneal angle measured in AP (anteroposterior) view. Used to assess flatfoot deformity/alignment.
1st Metatarsal–Talus Angle	**Lateral View**	Also called Meary's angle. Evaluates arch alignment and flatfoot/high arch condition.



Confidentiality & Data Protection

All data, images, measurements, X-ray images, AI models, system outputs, documents, and related materials used or generated throughout this development shall be treated as confidential and solely for the purpose of the AI Insole Foot Classification System development.

The developer shall:

- Maintain strict confidentiality of all provided datasets and project information.
- Not disclose, distribute, share, publish, or reuse any data or materials to third parties without prior written consent.
- Use the collected data only for model development, testing, validation, and deployment related to this project.
- Implement reasonable security measures to protect the data from unauthorized access, copying, leakage, or misuse.
- Ensure that any temporary or backup data storage is secured appropriately.
- Return or permanently delete the provided data upon project completion if requested by the project owner.

All intellectual property, datasets, trained models, reports, and system outputs developed under this project shall remain the property of the project owner unless otherwise agreed in writing.

Payment Terms

The payment for the development shall be made in two (2) phases:

1. **First Phase – 50%**
   - Payable upon project commencement and delivery of the first phase results/prototype/demo.
2. **Final Phase – Remaining 50%**
   - Payable upon full completion, final delivery, and acceptance of the system/project.

The staggered payment structure is intended to support progressive development, milestone verification, and successful project completion.